

SOFTWARE TEST DESCRIPTION (STD)

for the
Task Force Management System (TFMS)

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1 SCOPE

1.1. Identification

System Name : Task Force Monitoring System
System Abbreviation : TFMS
System Identification Number : TFMS-CODEForce

1.2. Overview of the System

Purpose of the Software System:

The purpose of this STD is to define testing activities of the TFMS, which replaces manual spreadsheet-based processes by managing lecturer task force assignments, workload fairness checks, and administrative reporting within the Faculty of Artificial Intelligence (FAI).

Purpose of the Software Documentation:

This Software Test Description (STD) document defines the test approach, test design, test cases, test procedures, and associated test data used to test the TFMS. The document specifies how testing activities are planned, designed, and executed to verify that the system satisfies both functional and non-functional requirements.



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1.3. Overview of the Documents

The Software Test Description (STD) the test strategy, test design, test cases, test procedures, and associated test data used necessary to perform formal qualification testing of a TFMS.

Chapter 1 Describes the scope, identification, system overview and document overview.

Chapter 2 Referenced documents, government documents and non-government documents.

Chapter 3 Describes the formal qualification test preparations.

Chapter 4 Describes the test cases, test procedures and related information associated with each formal qualification test.



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2 REFERENCED DOCUMENTS

The following documents, for which the exact is indicated, form part of the specification as far as everything specified hereafter is concerned. In the event of a discrepancy between the documents referred to here and the content of this specification, it is the content of the specification, which should be considered to be the background reference.

Copies of specifications, standards, drawings and publications requested by suppliers in contact with the specified supplying functions may be obtained by contacting the contracting agency or directly through the contracting office.

2.1 Government Documents

- [1] DoD-Std-2167A Defense System Software Development
- [2] DoD-Std-2168 Defense System Software Quality Program
- [3] MIL-Std-1521B Technical Reviews and Audits

2.2 Non-Government Documents

This section consists of the contractual documents and non-contractual documents.

2.2.1 Contractual documents

- [4] URS (MSE-FT31-2025) User Requirements Specification
- [5] SDP-TFMS-261225-V1 Software Development Plan
- [6] SRS-TFMS-140126-V1 Software Requirements Specification
- [7] SDD-TFMS-190126-V1 Software Design Documents
- [8] STR-TFMS-190126-V1 Software Test Report

2.2.2 Non-contractual document

- [9] Software Testing: A Craftsman's Approach, Paul C. Jorgensen (2018), CRC Press
- [10] Agile Testing: A Practical Guide for Testers and Agile Teams, Lisa Crispin & Janet Gregory (2009), Addison-Wesley
- [11] The Art of Software Testing, Glenford J. Myers (2011), John Wiley & Sons
- [12] ISO/IEC/IEEE 29119-2:2021 Software and Systems Engineering - Software Testing - Part 2: Test Process
- [13] ISO/IEC/IEEE 29119-3:2021 Software and Systems Engineering - Software Testing - Part 3: Test Documentation
- [14] ISO/IEC/IEEE 29119-4:2015 Software and Systems Engineering - Software Testing - Part 4: Test Techniques



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- [15] ISO/IEC 25010:2023 Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — Product quality model
 [16] https://glossary.istqb.org/en_US/search?term=&exact_matches_first=true

Glossary:

Term	Definition
Acceptance Criteria	The criteria that a work product must satisfy to be accepted by the stakeholders.
Functional Test	Functional testing is to verify the functionalities of the application modules and the system as stated in the requirements. Individual features within the specific system modules will be verified.
Non-Functional Test	Non-functional testing is the testing of a system for its non-functional requirements. Objective of the non-functional testing is to cover the quality attributes of the component or system such as performance test.
Performance Test	A test type to determine the performance efficiency of a component or system.
Regression Test	Regression testing is conducted after the first iteration of testing has been completed. Objective of regression testing is to find new defects in existing functionalities of a system after changes such as enhancements, patches or configuration changes have been made to the system.
Security Test	A test approach in which attacks are simulated and reactions are analyzed to determine the vulnerabilities of a component or system.
Test Case	A set of preconditions, inputs, actions (where applicable), expected results and postconditions, developed based on test conditions.
Test Condition	A testable aspect of a component or system that is intended to be tested.
Test Design	The activity that derives and specifies test cases from test conditions.
Test Item	A part of a test object used in the test process.
Test Model	A model describing testware that is used for testing a component or a system under test.



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Term	Definition
Test Procedure	A sequence of test cases in execution order and any associated actions that may be required to set up the initial preconditions and any wrap-up activities after execution.

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3 FORMAL QUALIFICATION TEST PREPARATIONS

3.1 Test Overview

Purpose:

The purpose of this STD is to define testing activities of the TFMS, which replaces manual spreadsheet-based processes by managing lecturer task force assignments, workload fairness checks, and administrative reporting within the Faculty of Artificial Intelligence (FAI).

Scope:

Testing scope is based on the Product Quality Model in ISO/IEC 25010:2023 including functional testing and non-functional testing such as performance and security testing for identified test items. It excludes integration with external finance or student-records systems.

Test Model versus Test Item:

Table 3.1 summarizes the relationship between the defined Test Model and the corresponding Test Items.

Table 3.1: Test Model versus Test Item

Test Model	Test Item Name
Authentication and Access Control	User Authentication & Login
	Password Reset
Staff and Task Force Management	Staff Management
	Task Force Management
	System Configuration
Assignment and Workload Management	Task Force Assignment (HOD)
Approval Workflow	Task Force Approval (PSM)
Lecturer Portfolio	Lecturer Portfolio
Executive Reporting	Executive Reports
Audit and Notification	Audit and Notification

Acceptance Criteria

Table 3.2 summarizes the acceptance criteria, which establish clear thresholds to determine whether a feature can be considered successfully tested.



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Table 3.2: Acceptance Criteria

Acceptance Criteria	
Functional test:	>95 % of functional requirements tested >95 % branch-coverage for critical modules. >95% overall pass with no high-severity test cases
Performance test:	90 percentile response times remain at or below the threshold for 50 virtual users ✓ Time ≤ 2 sec ✓ 5 sec < Time and < 10 sec ✓ Time > 10 sec Server Utilization < 80% Failure rate < 2%
Security test:	no critical or high-severity findings remain unresolved.

Test Deliverable:

- Specification test description
- Test Execution: Test Procedure/Test Script, Test Data.
- Test Data Sets: CSV and SQL scripts for seeding test scenarios.
- Traceability Matrix: Mapping of SRS requirements to test cases and procedures.
- Defect Reports: Jira tickets with reproduction steps, severity, and status.
- Test Report:
 - ✓ Functional Test Result (Test Summary by Test Item)
 - ✓ Test Execution Logs
 - ✓ Performance Test Result, if any
 - ✓ Security Test Result, if any

3.2 Testing Schedule

The testing will be conducted according to the following schedule:

Testing Activities	December 2025		January 2026				
	W3	W4	W1	W2	W3	W4	W5
Unit Testing							
Integration Testing							
System Testing							
Acceptance Testing							
Security Testing							
Performance Testing							

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Remark: 1 week = 5 working days

3.3 Preparation Procedure of the Test

The preparation procedure for testing the **TFMS** ensures that all necessary preconditions, test environments, and data are in place before executing test cases. The preparation steps include:

1. Test Environment Setup

- Deploy TFMS on a supported web server with access to the test database.
- Verify network connectivity and browser compatibility for all user roles (Admin, HOD, PSM, Lecturer and Management).

2. Test Data Preparation

- Create sample user accounts for each role with valid and invalid credentials.
- Populate the system with representative staff, task force, and workload records to simulate realistic scenarios.

3. Access and Permissions Verification

- Ensure role-based access control is configured correctly for each test account.
- Confirm that administrative, assignment, and reporting functionalities are accessible according to user roles.

4. Test Tools Configuration

- Set up automated testing tools such as Selenium WebDriver, if automation is used.
- Configure logging and monitoring tools to capture test execution results and system responses, if any.

5. Validation of Preconditions

- Confirm TFMS main functionalities are operational.
- Verify initial system state aligns with test scenarios, including account statuses and task force assignments.

3.4 Test Environment Requirements

The contents for test environment requirements are as follows:

- Unique identifier
- Test environment item
- Description
- Responsibility
- Period needed



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Test Environment Item:

Test execution will be conducted across three distinct environments and configurations, as summarized in Tables 3.2 and 3.3

Table 3.3: Test Environment

Description \ Environment	Development	Staging	Production
Platform	Local development machines	PythonAnywhere (Free Tier)	To be determined
Tools	Git, Visual Studio Code	PythonAnywhere Console, Manual Testing (Browser)	
Operating System	macOS	Linux (Ubuntu - PythonAnywhere)	
Database	SQLite		
Backend	Django 4.2.27, Python 3.10+		
Frontend	Django Templates		

Table 3.4: Test Activities versus Test Environment

Environment	Testing Type/Level
Development	Unit Test/Component/Component Integration/System Test
Staging	Acceptance Test
Production	Non-functional Test (performance and security test)

3.5 Test Strategy

The testing strategy adopts a **risk-driven scope selection**, where only **Very High** and **High** priority test cases are identified, designed, and executed. **Very High priority test cases** are those whose failure would critically impact core system functionality, security, or access control. **High priority test cases** are those whose failure would significantly affect workflow integrity or user roles, but without immediate catastrophic impact.

These priorities are determined based on the potential impact to core system functionality, security, access control, and cross-role workflow integrity. Lower-priority test cases are intentionally deferred, as they present limited risk to overall system reliability within the current testing scope.

In addition to scripted test execution, the strategy incorporates **unscripted exploratory testing to uncover defects** that may not be captured through

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predefined test cases. This exploratory approach enables testers to investigate system behavior, validate assumptions, and identify unexpected interoperability issues, thereby complementing the risk-based coverage with real-world insights.

3.6 Test Model Specification

This phase sets the foundation by defining the system's expected behaviours, scenarios, and criteria. It acts as the blueprint, a logical, architectural model that pinpoints where and how the system should be evaluated. A test model is a structured representation of a test item, highlighting the attributes that determine the needed test coverage. Combining the test model with the required coverage criteria allows for deriving specific test coverage items.

The **test design** is derived from defined **test models and test conditions**, which represent the expected system behaviours, execution flows, and operational states of each test item.

A full specification may include unique identifier, objective, priority, test strategy extract, test model, traceability. In practice, this project focuses on the most significant elements such as objective, test model, and traceability to ensure clarity and requirement coverage. Other elements are acknowledged but not detailed.

3.7 Test Case Specification

Test cases are systematically derived using established test design techniques, including **Equivalence Partitioning (EP)**, **Boundary Value Analysis (BVA)** and **Use Case Testing** to ensure efficient and systematic input coverage at the unit test and component test level. Alongside **positive and negative testing** to validate both expected and erroneous system behaviour. Additionally, **main flow and alternative flow testing** is employed to verify normal operations as well as exception handling paths. Where applicable, multiple test cases may be executed within a single test scenario to efficiently validate end-to-end functionality.

Test cases are executed either individually or a collection within a test scenario, depending on the structure of the underlying requirements and test conditions. Multiple test cases may be executed under a single test scenario when the related requirements form an inseparable end-to-end execution flow. In such cases, the scenario validates multiple interconnected system behaviours while maintaining traceability to individual SRS requirements through defined test conditions. Conversely, a single requirement may result in multiple test cases where different operational conditions, data partitions, or boundary values must be evaluated.

A full specification may include unique identifier, description, priority, traceability for test coverage item. Test case may include unique identifier, objective, priority, traceability, preconditions, inputs, expected results. In practice, this project focuses on the most significant elements such as identifier, description, priority, traceability and preconditions, inputs, expected results to ensure clarity and requirement coverage. Other elements are acknowledged but not detailed.



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The following table provides descriptions of the applied test design techniques.

Table 3.5: Test Techniques Description

Test Techniques	Description
Equivalence Partitioning (EP)	A black-box test technique in which test conditions are equivalence partitions exercised by one representative member of each partition.
Boundary Value Analysis (BVA)	A black-box test technique in which test cases are designed based on boundary values.
Positive Testing	A test type focuses on valid inputs and expected conditions to confirm correct functionality.
Negative testing	A test type in which a component or system is used in a way that it is not intended.
Use Case Testing	A black-box test technique in which test cases are designed to exercise use case behaviors.

3.8 Test Procedure Specification

The test procedure specification defines the **ordered execution of TFMS test cases**, including all associated actions required to establish **role-based preconditions** (e.g., Admin, HOD, PSM, Lecturer access), prepare relevant **task force and staff data**, execute the defined test steps, and complete **post-execution wrap-up activities** such as session termination, data rollback, and test result documentation.

A full specification may include unique identifier, objective, priority, start up, ordered test cases, relationship to other procedures, stop and wrap up. In practice, this project focuses on the most significant elements to ensure clarity and requirement coverage. Other elements are acknowledged but not detailed.

Refer to Section 4: Formal Qualification Test Descriptions for the list of test procedure.

3.9 Test Data Requirements

A full specification may include unique identifier, description, responsibility, period needed, resetting needs, archiving or disposal. In practice, this project focuses on the most significant elements to ensure clarity and requirement coverage. Other elements are acknowledged but not detailed.

General description for test data requirements:



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- Test database seeded with representative TFMS records, including staff profiles, roles (Admin, HOD, PSM, Lecturer), task force definitions, workload values, and approval states to cover all major operational scenarios.
- Anonymized sample data sets reflecting realistic TFMS usage, ensuring no real personnel or institutional data is exposed during testing.
- Scripted data loaders to populate staff accounts, task force memberships, workload allocations, approval submissions, feedback entries, and system notification templates.
- Performance testing data prepared in .csv format, representing large volumes of staff records, task force assignments, and historical workload data to simulate peak operational conditions.
- Security testing data, including role-based user credentials, authentication tokens, and predefined critical access paths, to validate access control, session management, and authorization enforcement across TFMS modules.

3.10 Other Pre-test Preparations

- Initialize test database with baseline data
- Configure test user accounts
- Prepare test files for upload scenarios
- Set up test email accounts



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4 FORMAL QUALIFICATION TEST DESCRIPTIONS

This section presents the formal qualification test descriptions for the TFMS, focusing on Very High and High priority test cases derived through risk-based analysis. The tests are designed to validate critical system functionality, ensuring traceability between requirements, test conditions, and test cases. Refer Annex 1: Requirement Traceability Matrix.

4.1 Test Qualification Summary

Table 3.5 summarizes the mapping between the defined Test Models and their corresponding Test Items. It illustrates how each model element is linked to specific test items together with their associated SRS coverage, thereby ensuring traceability and systematic validation of requirements.

Table 3.6: Test Qualification Summary

Test Model ID	Test Model	Test Item ID	Test Item Name	SRS Coverage
TM-AUTH-01	Authentication and Access Control	TI-AUTH-01	User Authentication and Login	SRS_REQ_101-109
		TI-AUTH-02	Password Reset	SRS_REQ_110-119
TM-ADMIN-01	Staff and Task Force Management	TI-ADMIN-01	Staff Management	SRS_REQ_201-207
		TI-ADMIN-02	Task Force Management	SRS_REQ_208-209, SRS_REQ_213-214
		TI-ADMIN-03	System Configuration	SRS_REQ_210
TM-HOD-01	Assignment and Workload Management	TI-HOD-01	Task Force Assignment (HOD)	SRS_REQ_301-312
TM-PSM-01	Approval Workflow	TI-PSM-01	Task Force Approval (PSM)	SRS_REQ_401-409
TM-LECT-01	Lecturer Portfolio	TI-LECT-01	Lecturer Portfolio	SRS_REQ_501-509
TM-MGMT-01	Executive Reporting	TI-MGMT-01	Executive Reports	SRS_REQ_601-606
TM-AUD-01	Audit and Notification	TI-AUD-01	Audit & Notification	SRS_REQ_211-212 SRS_REQ_407, SRS_REQ_506, SRS_REQ_606



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4.2 Formal Qualification Test Cases and Procedures Overview

The requirements in the SRS_REQ_100, which focus on authentication and access control, are addressed through Test Model TM-AUTH-01. This model refers to two test items: TI-AUTH-01 (User Authentication and Login), covering SRS_REQ_101–109, and TI-AUTH-02 (Password Reset), covering SRS_REQ_110–119. Together, these items ensure secure login validation and password recovery are comprehensively tested.

Additionally, TI-AUD-01, Audit & Notification, cross-references SRS_REQ_211–212, SRS_REQ_407, SRS_REQ_506, and SRS_REQ_606. This test item ensures audit trail completeness and notification reliability across the system.

4.2.1 Test Item: User Authentication and Login

Test Item ID: TI-AUTH-01

Test Item Name: User Authentication and Login

Priority: Very High

Test Level: System / Acceptance

4.2.1.1 [TI-AUTH-01] Test Cases and Procedures Overview

Test Item ID: TI-AUTH-01

Test Case ID: TC-01-01

Test Case: Navigate to Login Page and Display Form

Priority: Very High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: User accesses TFMS login page and the login form is displayed correctly

SRS Traceability: SRS_REQ_101, SRS_REQ_102

Test Objective:

To verify that the TFMS login page is accessible and that the login form appears as expected.

Preconditions:



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- TFMS application is accessible via network
- Browser is supported and open

Test Data:

- None required (page access only)

Test Procedure:

1. User opens the TFMS URL in a supported web browser.
2. Observe and verify that the login form appears with:
 - Username input field
 - Password input field
 - "Login" button
 - "Forgot Password?" link

Expected Results:

- TFMS login page loads successfully
- Login form is visible with all required fields and buttons

Postconditions:

- User remains on the login page
- No session is created

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Test Case ID: TC-01-02

Test Case: Valid Login with Correct Credentials

Priority: Very High

Test Technique: Use Case Testing, Equivalence Partitioning (valid credentials), Positive Testing

Test Scenario: User logs in successfully with valid username and password

SRS Traceability: SRS_REQ_103, SRS_REQ_104, SRS_REQ_105

Test Objective:

To verify that TFMS accepts valid user credentials and successfully validates them when the Login action is performed.

Preconditions:

- TFMS login page is displayed
- User account exists and is active
- User account is not locked

Test Data:

- Valid username
- Valid password

Test Procedure:

1. User enters a valid username in the username field.
2. User enters a valid password in the password field.
3. User clicks the "Login" button.
4. TFMS validates the entered credentials.

Expected Results:

- No validation error message is displayed
- Credentials are accepted by the system



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Postconditions:

- User authentication is successful
- User proceeds to the next authentication step (session creation / dashboard display handled in subsequent test cases)

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Test Case: Invalid Login with Incorrect Credentials

Priority: Very High

Test Technique: Use Case Testing, Equivalence Partitioning (invalid credentials), Negative Testing

Test Scenario: User attempts to log in using invalid username and/or password

SRS Traceability: SRS_REQ_103, SRS_REQ_104, SRS_REQ_105

Test Objective:

To verify that TFMS rejects invalid user credentials and fails authentication when the Login action is performed.

Preconditions:

- TFMS login page is displayed
- User account exists but credentials entered are incorrect
- User account is not locked

Postconditions:

- No user session is created
- User is not logged into TFMS

Child Test Cases:

Test Case ID: TC-01-03-01

- **Scenario:** User enters invalid username and invalid password.
- **Test Data:** Invalid username and invalid password combination.

- **Test Procedure:**

1. Enter invalid username and invalid password.
2. Click "Login."
3. System validates credentials.



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- **Expected Results:**

- Authentication fails.
- Error message displayed: "Invalid username or password."
- User remains on login page.

Test Case ID: TC-01-03-02

- **Scenario:** User enters invalid username only.
- **Test Data:** Invalid username with valid password.

- **Test Procedure:**

1. Enter invalid username with valid password.
2. Click "Login."
3. System validates credentials.

- **Expected Results:**

- Authentication fails.
- Error message displayed: "Invalid username."
- User remains on login page.

Test Case ID: TC-01-03-03

- **Scenario:** User enters invalid password only.
- **Test Data:** Valid username with invalid password.

- **Test Procedure:**

1. Enter valid username with invalid password.
2. Click "Login."
3. System validates credentials.

- **Expected Results:**

- Authentication fails.
- Error message displayed: "Invalid password."

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	19 /120

- User remains on login page.

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DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	20 /120

Test Case: Login Attempt with Empty Username and/or Password

Priority: Very High

Test Technique: Use Case Testing, Boundary Value Analysis (empty input), Negative Testing

Test Scenario: User attempts to log in without providing mandatory login credentials

SRS Traceability: SRS_REQ_103, SRS_REQ_104, SRS_REQ_105

Test Objective:

To verify that TFMS detects missing mandatory login fields and prevents authentication when the Login action is performed.

Preconditions:

- TFMS login page is displayed
- User account exists
- User account is not locked

Postconditions:

- No user session is created
- No authentication state is established

Child Test Cases:

Test Case ID: TC-01-04-01

- **Scenario:** User enters empty username and empty password.
- **Test Data:** Username = "" and Password = ""

• **Procedure:**

1. Leave both fields empty.
2. Click "Login."
3. System performs input validation.

• **Expected Results:**



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Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	1 / 120

- Mandatory field warning messages displayed for both fields.
- Authentication fails.
- User remains on login page.

Test Case ID: TC-01-04-02

- **Scenario:** User enters empty username only.
- **Test Data:** Username = "" ; Password = valid.

- **Procedure:**

1. Leave username empty, enter valid password.
2. Click "Login."
3. System performs input validation.

- **Expected Results:**

- Mandatory field warning message displayed for username.
- Authentication fails.
- User remains on login page.

Test Case ID: TC-01-04-03

- **Scenario:** User enters empty password only
- **Test Data:** Username = valid; Password = ""

- **Procedure:**

1. Enter valid username, leave password empty.
2. Click "Login."
3. System performs input validation.

- **Expected Results:**

- Mandatory field warning messages displayed for both fields.
- Authentication fails.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	2 /120

- User remains on login page.

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SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	3 /120

Test Case ID: TC-01-05

Test Case: Login Attempt with Locked Account

Priority: Very High

Test Technique: Use Case Testing, State Transition Testing, Negative Testing

Test Scenario: User attempts to log in using an account that is locked

SRS Traceability: SRS_REQ_106

Test Objective:

To verify that TFMS detects a locked user account and prevents login by displaying an appropriate lock message.

Preconditions:

- TFMS login page is displayed
- User account exists
- User account is locked

Test Data:

- Valid username and password for a locked account

Test Procedure:

1. User enters valid credentials for a locked account.
2. User clicks the "Login" button.
3. TFMS checks the account lock status.

Expected Results:

- Login is rejected
- Account locked message is displayed
- User remains on the login page

Postconditions:

- No user session is created



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	1 / 120

Test Case ID: TC-01-06

Test Case: First-Time Login Requires Password Change

Priority: Very High

Test Technique: Use Case Testing, Decision Table Testing

Test Scenario: User logs in for the first time and is required to change the initial password

SRS Traceability: SRS_REQ_107

Test Objective:

To verify that TFMS detects a first-time login and prompts the user to change the initial password.

Preconditions:

- TFMS login page is displayed
- User account exists and is active
- User is logging in for the first time

Test Data:

- Valid first-time username and password

Test Procedure:

1. User enters valid credentials.
2. User clicks the "Login" button.
3. TFMS identifies the login as a first-time login.

Expected Results:

- User is prompted to change the password
- Access to the dashboard is not granted until password is updated

Postconditions:

- User is redirected to password change page



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	2 /120

Test Case ID: TC-01-07

Test Case: User Session Creation After Successful Login

Priority: Very High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: TFMS creates a session for a successfully authenticated user

SRS Traceability: SRS_REQ_108

Test Objective:

To verify that TFMS establishes a user session after successful authentication.

Preconditions:

- User credentials are valid
- User account is active and not locked
- User is not required to change password

Test Data:

- Valid username and password

Test Procedure:

1. User completes a successful login.
2. TFMS creates a user session.

Expected Results:

- User session is successfully created
- Session information is maintained by the system

Postconditions:

- User is authenticated within TFMS



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	3 /120

Test Case ID: TC-01-08

Test Case: Display Role-Appropriate Dashboard After Login

Priority: Very High

Test Technique: Use Case Testing, Role-Based Access Testing

Test Scenario: User is presented with the correct dashboard based on assigned role

SRS Traceability: SRS_REQ_109

Test Objective:

To verify that TFMS displays the correct dashboard according to the user's assigned role after login.

Preconditions:

- User is authenticated
- User session is active
- User role is assigned (Admin / HOD / PSM / Lecturer)

Test Data:

- Valid user credentials with assigned role

Test Procedure:

1. User completes login successfully.
2. TFMS identifies the user role.
3. TFMS displays the corresponding dashboard.

Expected Results:

- Role-appropriate dashboard is displayed
- Unauthorized dashboards are not accessible

Postconditions:

- User remains logged in with appropriate access



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	4 /120

Test Case ID: TC-01-09

Test Case: Account Locked After Multiple Failed Login Attempts

Priority: Very High

Test Technique: Use Case Testing, State Transition Testing, Negative Testing

Test Scenario: User account is locked after exceeding the maximum number of failed login attempts

SRS Traceability: SRS_REQ_115, SRS_REQ_116, SRS_REQ_117,
SRS_REQ_118, SRS_REQ_119

Test Objective:

To verify that TFMS increments failed login attempts, locks the user account after the defined threshold, and displays the appropriate lock notification.

Preconditions:

- TFMS login page is displayed
- User account exists and is active
- User account is not initially locked

Test Data:

- Valid username
- Invalid password

Test Procedure:

1. User enters a valid username with an invalid password.
2. User clicks the "Login" button.
3. TFMS increments the failed login attempt counter.
4. Steps 1–3 are repeated until the maximum allowed failed attempts is reached.
5. User attempts to log in again after the threshold is exceeded.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	5 /120

Expected Results:

- Failed login attempt counter is incremented after each failed attempt
- User account is locked after the maximum number of failed attempts
- Account locked message is displayed
- Further login attempts are blocked

Postconditions:

- User account remains locked
- No user session is created

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SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	6 /120

4.2.2 Test Item: Password Reset

Test Item ID: TI-AUTH-02

Test Item Name: Reset Password

Priority: Very High

Test Level: System / Acceptance

4.2.2.1 [TI-AUTH-02] Test Cases and Procedures Overview

Test Case ID: TC-02-01

Test Case: Initiate Password Reset via "Forgot Password?"

Priority: Very High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: User requests a password reset and receives a reset link via registered email

SRS Traceability: SRS_REQ_110, SRS_REQ_111

Test Objective:

To verify that TFMS allows a user to initiate a password reset and sends a reset link to the registered email address.

Preconditions:

- TFMS login page is displayed
- User account exists with a registered email address

Test Data:

- Valid registered email address

Test Procedure:

1. User clicks the "Forgot Password?" link on the login page.
2. TFMS displays the password reset request form.
3. User enters a registered email address.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	7 /120

4. User submits the reset request.

Expected Results:

- TFMS validates the email address
- Password reset link is generated
- Reset link is sent to the user's registered email

Postconditions:

- User account password remains unchanged
- Reset link is active and pending use

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	8 /120

Test Case ID: TC-02-02**Test Case:** Enforce Password Change for Initial or Unmodified Password**Priority:** Very High**Test Technique:** Use Case Testing, Conditional Testing**Test Scenario:** TFMS detects an unmodified initial password and forces password change**SRS Traceability:** SRS_REQ_112, SRS_REQ_113**Test Objective:**

To verify that TFMS detects the use of an initial or unmodified password and requires the user to change it before proceeding.

Preconditions:

- User account is created with an initial/default password
- User logs in successfully using the initial password

Test Data:

- Valid username
- Initial/default password

Test Procedure:

1. User logs in using the initial/default password.
2. TFMS evaluates the password status.
3. TFMS displays the password change form.

Expected Results:

- TFMS prevents access to the dashboard
- Password change form is displayed
- User is required to set a new password

Postconditions:

- User session remains restricted
- Password is not updated until user submits a valid new password

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	9 /120

Test Case ID: TC-02-03**Test Case:** Update Password Successfully**Priority:** Very High**Test Technique:** Use Case Testing, Positive Testing**Test Scenario:** User sets a new password and the system updates it successfully**SRS Traceability:** SRS_REQ_114**Test Objective:**

To verify that TFMS allows the user to update their password successfully after validation.

Preconditions:

- Password change form is displayed
- User is authenticated or has accessed the reset link

Test Data:

- Valid new password meeting complexity rules

Test Procedure:

1. User enters a new password.
2. User confirms the new password.
3. User submits the password change form.

Expected Results:

- TFMS validates password complexity rules
- Password is updated successfully
- Confirmation message is displayed

Postconditions:

- New password is stored securely
- User may proceed to login or dashboard

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	10 /120

The requirements in the SRS_REQ_200 range, which cover staff and task force management, are addressed through Test Model TM-ADMIN-01. This model refers to three test items: TI-ADMIN-01 (Staff Management), covering SRS_REQ_201–207; TI-ADMIN-02 (Task Force Management), covering SRS_REQ_208–209 and SRS_REQ_213–214; and TI-ADMIN-03 (System Configuration), covering SRS_REQ_210. These items validate administrative functions and configuration controls.

4.2.3 Test Item: Staff Management

Test Item ID: TI-ADMIN-01

Test Item Name: Staff Management

Priority: Very High

Test Level: System / Acceptance

4.2.3.1 [TI-ADMIN-01] Test Cases and Procedures Overview

Test Item ID: TI-ADM-01

Test Case ID: TC-03-01

Test case: Accessing the employee management module

Priority: Very High

Test Technique: Use Case Testing, Positive testing

Test Scenario: Administrator access to employee management.

SRS Traceability: SRS_REQ_201

Test Objective:

Verify that the administrator user can successfully access the employee management module from the TFMS dashboard.

Preconditions:

- The administrator is logged into TFMS.
- The administrator role has the appropriate permissions to access employee management.

Test Data:

- A valid administrator username and password.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	11 /120

Test Procedure:

1. The administrator logs in to TFMS using valid credentials.
2. After logging in, the administrator is taken to the dashboard interface.
3. Locate and click "Employee Management" in the dashboard.
4. Observe the page loading and the displayed interface.

Expected Results:

- The employee management module page loaded successfully.
- The interface displays a list of existing employees.
- The interface includes employee information (employee ID, username, name, role, status), operations (edit), and an "Add New Employee" button.

Postconditions:

- The administrator remains on the employee management module interface.
- No changes have been made to any employee records.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	12 /120

Test Case ID: TC-03-02-01

Test case: Add new employees and generate system IDs

Priority: Very High

Test Technique: Use Case Testing, Positive testing

Test Scenario: When an administrator adds an employee record, TFMS automatically generates an employee ID when the employee is created.

SRS Traceability: SRS_REQ_202, SRS_REQ_203

Test Objective:

When the administrator fills out the "Add New Employee" form, the system successfully creates a new employee record and automatically generates a unique employee ID.

Preconditions:

- The administrator is logged into TFMS and on the employee management page.
- The "Add New Employee" function is accessible.

Test Data:

- New Employee Details:
 - Username: lecturer1
 - First Name: Test
 - Last Name: One
 - Email: OneTest@graduate.utm.my
 - Role: Lecturer

Test Procedure:

1. Click the "Add New Employee" button.
2. Enter a unique username in the "Username" field.
3. Enter the "First Name" and "Last Name" fields.
4. Enter a valid email address in the "Email" field.
5. Select a "Role" from the drop-down menu.
6. Select a "Department" from the drop-down menu.
7. Click the "Save Employee Information" button.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	13 /120

8. Observe the system response message.
9. After successful creation, navigate to the employee list.
10. View the "Employee ID" field in the list and record its value.

Expected Results:

- The system confirms "Staff created. Email sent to OneTest@graduate.utm.my."
- The new employee appears in the employee list.
- A unique "Employee ID" is displayed in the employee list, and this ID is not the one the administrator entered in the creation form.
- The "Employee ID" follows a consistent system-defined format and is unique within the system.

Postconditions:

- New employee records are saved in the system with automatically generated employee IDs.
- Administrators remain in the employee management module.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	14 /120

Test Case ID: TC-03-02-02

Test case: Admin edits staff records

Priority: Very High

Test Technique: Use Case Testing, Positive testing

Test Scenario: The administrator edits employee records.

SRS Traceability: SRS_REQ_202, SRS_REQ_203

Test Objective:

Verify that the administrator can successfully modify the details of existing employees and that the changes are correctly saved.

Preconditions:

- The administrator is on the employee management page.
- At least one employee record exists and is visible in the list.

Test Data:

- Existing employee usernames.
- Update: Role changed from HOD to PSM.

Test Procedure:

1. Locate the target employee in the employee list.
2. Click the "Edit" button associated with that employee.
3. In the edit form, select PSM from the "Role" dropdown.
4. Click the "Update Employee" button.
5. Observe the system's confirmation message.
6. Verify that the changes are reflected in the employee list.

Expected Results:

- The employee update was successful.
- The employee list displays the employee's updated role.
- The employee ID and other unedited fields remain unchanged.

Postconditions:

- Employee records are updated to PSM.
- The administrator remains in the employee management module.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	15 /120

Test Case ID: TC-03-03

Test case: Assign user roles to employees

Priority: Very High

Test Technique: Use Case Testing, Positive testing

Test Scenario: Administrators assign roles to users.

SRS Traceability: SRS_REQ_204

Test Objective:

Verification administrators can assign specific system roles to employees during the creation or editing process.

Preconditions:

- The administrator is adding a new employee or editing an existing employee.
- The role selection dropdown menu is available in the employee form.

Test Data:

- Employee details (new or existing).
- Role to be assigned.

Test Procedure:

1. For new employees: Navigate to the "Add New Employee" form.
2. For existing employees: Navigate to the employee's edit form.
3. In the form, find the "Role" dropdown selection field.
4. Select the desired role from the dropdown list.
5. If creating a new employee, complete the other required fields.
6. Click the "Save" button.
7. After saving, verify the employee's assigned role.

Expected Results:

- The system successfully saved the employee information.
- The employee's record displays the newly assigned role in the employee list.
- This employee should then have the access permissions associated with the Department Head role.



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SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	16 /120

Postconditions:

- Employees are associated with designated user roles.
- Role-based permissions apply to the employee's account.

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	DOCUMENT IDENTIFICATION				
	SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
	Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	17 /120

Test Case ID: TC-03-04

Test case: The system creates user accounts with temporary passwords.

Priority: Very High

Test Technique: Use Case Testing, Positive testing

Test Scenario: TFMS creates an account with a temporary password.

SRS Traceability: SRS_REQ_205

Test Objective:

The system automatically creates a new user account for them and protects it with a system-generated temporary password, which must be changed upon first login.

Preconditions:

- The administrator is adding a new employee via the "Add New Employee" form.
- This new employee does not yet have a user account.

Test Data:

- New employee details, including a unique username and selected role.

Test Procedure:

1. Complete the "Add New Employee" form using a unique username, name, email address, selected role, and department.
2. Click "Save Employee".
3. Observe the screen for any messages regarding account creation or password settings.
4. Try logging in as the new employee:
 - Log out of the administrator account.
 - On the TFMS login page, enter the username of the newly created employee.
 - Try logging in without a password or using a common default password (if known). Observe the system response.
 - Alternatively, check if the system sends a "welcome email" containing a temporary password or setup link to the provided email address.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	1 /120

Expected Results:

- Employee record created successfully.
- The system does not allow login with blank passwords or common default passwords.
- The system will either:
 - send a temporary password via email for first-time login, or
 - force a password reset on the first login attempt, prompting the user to set a new password immediately.
- The new employee account status is marked as requiring a password change.

Postconditions:

- User accounts have been created for new employees.
- Accounts are protected through an initial password process managed by the system.
- Employees must change their passwords upon their first successful login.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	2 /120

Test Case ID: TC-03-05

Test case: Administrator resets password and unlocks account

Priority: Very High

Test Technique: Use Case Testing, Positive testing

Test Scenario: The administrator resets the password and unlocks the account.

SRS Traceability: SRS_REQ_206

Test Objective:

The administrator is authorized to reset passwords for employee accounts that have been locked or whose passwords have been forgotten, and to verify that the account can be logged in normally after the reset.

Preconditions:

- The administrator is logged into TFMS.
- The target employee account exists, and its status is "locked" or the password is known and needs to be reset.
- The administrator has user account management privileges.

Test Data:

- The target employee's username.
- The employee's registered email address (used to receive the reset link).

Test Procedure:

1. Navigate to the Account Management section under "Employee Management".
2. Locate the target employee account in the user list.
3. Click the "Edit" button next to the account.
4. On the Edit Staff Member page, click the "RESET PASSWORD" button.
5. Observe the system response message, noting any prompts for temporary password generation or a reset link.
6. Verify the reset effect:
 - Try logging in using the target employee's username.
 - If the system displays "Account locked", the unlocking operation failed.
 - If the system sent a reset link, click the link and set a new password.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	3 /120

- Try logging in using the newly set password.

Expected Results:

- The system confirmed "Password reset successful."
- The target employee's account status changed from "Locked" to "Active."
- The employee can successfully log in to the system using the password set during the reset process.
- The relevant operations were recorded in the audit log.

Postconditions:

- The target employee's account password has been reset, and the account has been unlocked.
- The employee can now access the system using the new password.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	4 /120

Test Case ID: TC-03-06

Test case: Administrator deactivates employee

Priority: Very High

Test Technique: Use Case Testing, Negative testing

Test Scenario: The administrator deactivates the employee.

SRS Traceability: SRS_REQ_207

Test Objective:

The administrator can verify that an active employee's status can be changed to "disabled," and that after being disabled, the employee cannot log in to the system and is not visible when new tasks are assigned.

Preconditions:

- The administrator is logged into TFMS.
- The target employee account exists and is in an "active" state.
- This employee currently has no ongoing critical tasks or task group responsibilities.

Test Data:

- The name or username of the target active employee.

Test Procedure:

1. Navigate to the "Employee Management" module.
2. Find the target employee in the employee list.
3. Click the "Edit" button next to the employee's record.
4. Navigate the Edit Staff Member page and click the "DEACTIVATE" button.
5. Confirm the deactivation operation and provide a reason (required).
6. Observe the system confirmation message.
7. Verify the deactivation effect:
 - Check if the employee's status has changed to "Inactive" in the employee list.
 - Try logging into TFMS with the employee's username and password.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	5 /120

- Navigate to the task group assignment interface and try assigning a new task to the employee to verify if they are no longer in the available employee list.

Expected Results:

- The system confirms the message "Employee successfully deactivated".
- The employee's status in the employee list or details page is updated to "Inactive".
- This employee can no longer log in to the system using their credentials.
- This employee will not appear in the list of available personnel when assigning new tasks.
- Their task history should be retained.

Postconditions:

- The target employee's status has been changed to "Inactive".
- The employee's login privileges have been revoked.
- The administrator remains in the employee management interface.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	6 /120

4.2.4 Test Item: Task Force Management

Test Item ID: TI-ADMIN-02

Test Item Name: Task Force Management

Priority: Very High

Test Level: System / Acceptance

4.2.4.1 [TI-ADMIN-02] Test Cases and Procedures Overview

Test Case ID: TC-04-01

Test Case: Administrator Creates Task Group and System Generates Task Group ID

Priority: Very High

Testing Techniques: Use Case Testing, Positive testing

Test Scenario: Administrator creates a task group, and TFMS automatically generates a task group ID.

SRS Traceability: SRS_REQ_208, SRS_REQ_209

Test Objective:

Verify that the administrator can create new task groups, and that the system automatically generates a unique task group ID upon creation.

Prerequisites:

- The administrator is logged into TFMS.
- The administrator is located in "Task Force Management".
- The "CREATE TASK FORCE" function is available.

Test Data:

- New Task Group Information:
 - Name: Test1
 - Departments: Software Engineering
 - Description: test1
 - Status: active



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	7 /120

- Weightage: 5

Test Procedure:

1. Click the "CREATE TASK FORCE" button.
2. Fill in the required information in the creation form: Name, Departments, Description, Status, and Weightage.
3. Note: There should be no "Task Group ID" input field in the form.
4. Click the "CREATE TASK FORCE" button.
5. Observe the system confirmation message.
6. In the task group list or the details page of the newly created task, find and record the value of the "Task Group ID" field.

Expected Result:

- The system confirms with a "Task group created successfully" message.
- The new task group appears in the task group list.
- The list or details page displays a unique "Task Group ID" automatically generated by the system (e.g., TF-2025-001).
- This ID was not entered by the administrator in the creation form.

Postcondition:

- A new task group record is created with a system-generated task group ID. The task group is in an "Active" or "Inactive" state.

**DOCUMENT IDENTIFICATION**

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Test Case ID: TC-07-01

Test Case: Administrator Deactivates Employee

Priority: High

Testing Techniques: Use Case Testing, Negative testing

Test Scenario: Admin deactivates staff members

SRS Traceability: SRS_REQ_213

Test Objective:

Verify that an administrator can deactivate an employee.

Prerequisites:

- The administrator is logged into TFMS.
- The target employee account exists and its status is "Active".

Test Data:

- The name of the target active employee.

Test Procedure:

1. Navigate to the "Employee Management" module.
2. Find the target employee in the employee list.
3. Click "Edit", then click the "DEACTIVATE" button.
4. When the system pops up a confirmation dialog box, requiring "Justification" to confirm the deactivation operation.
5. Observe the system confirmation message and check the employee's status in the list.

Expected Result:

- The employee's status column in the employee list updates to "Inactive".
- The employee is no longer appearing in the "Active Employees" filter list.

Postconditions:

- The target employee's status has changed to "Inactive".
- The employee's login permissions have been revoked.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	9 /120

Test Case ID: TC-07-02

Test Case: Administrator Deactivates Task Group

Priority: High

Testing Techniques: Use Case Testing, Negative testing

Test Scenario: Admin deactivates task forces.

SRS Traceability: SRS_REQ_214

Test Objective:

Verify that an administrator can change the status of an active task group to "Deactivated." After deactivation, the task group should no longer be assigned new members or appear in the active list.

Prerequisites:

- The administrator is logged into TFMS.
- There is an active task group.
- The "Task Force Management" module is accessible.

Test Data:

- The name of an active task group.

Test Procedure:

1. Navigate to the "Task Force Management" module.
2. Find the target task group in the task group list.
3. Click the "Edit" button for the task group.
4. In the "Status" dropdown menu, select "Inactive".
5. Click "Save Changes" to return to the "Task Force Management" module.
6. Check the status of the task group in the task group list and try to see if it is available when assigning members.



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SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
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Expected Result:

- The system confirms the "Task group is deactivated" message.
- The task group's status is updated to "Inactive" in the task group list or details page.
- When assigning task group members in HOD, the deactivated task group should not appear in the list of available task groups.
- Its historical members and records should be retained.

Postcondition:

- The target task group's status changes to "Inactive".
- This task group is no longer used for new job assignments.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	11 /120

4.2.5 Test Item: System Configuration

Test Item ID: TI-ADMIN-03

Test Item Name: System Configuration

Priority: Very High

Test Level: System / Acceptance

4.2.5.1 [TI-ADMIN-03] Test Cases and Procedures Overview

Test Case ID: TC-05-01

Test Case: Administrator Configures TFMS Parameters

Priority: High

Testing Techniques: Case Testing, Positive testing

Test Scenario: Administrator configures TFMS parameters.

SRS Traceability: SRS_REQ_210

Test Objective:

Verify that the administrator can access the system configuration interface and successfully modify and save system-level parameters.

Prerequisites:

- The administrator is logged into TFMS.
- The administrator has system configuration permissions.
- The "System Configuration" module is accessible.

Test Data:

- Configuration item to be modified: Change "Weightage" from 15 to 18.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	12 /120

Test Procedure:

1. Navigate to the "Task Force Management" module.
2. Find the department to be modified and click "Modify".
3. Change the value of the target parameter from the original value to the new value (change "Weightage" from 15 to 18).
4. Click the "Save Changes" button.
5. Observe the system confirmation message.
6. Refresh the page and go to the relevant department to verify if the changes have been persisted.

Expected Result:

- The system confirms the message "Configuration successfully updated".
- The modified configuration item displays the newly set value (Weightage is displayed as 18).
- The changes take effect and may affect subsequent workload calculations or session filtering logic.

Postconditions:

- System parameters have been updated to the new values.
- Related calculations or business rules will be executed based on the new parameters.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	13 /120

The requirements in the SRS_REQ_300 range, which emphasize assignment and workload management, are addressed through Test Model TM-HOD-01. This model refers to TI-HOD-01, Task Force Assignment by HOD, covering SRS_REQ_301-312, ensuring workload distribution and assignment processes are properly validated.

4.2.6 Test Item: Task Force Assignment (HOD)

Test Item ID: TI-HOD-01

Test Item Name: Task Force Assignment (HOD)

Priority: Very High

Test Level: System / Acceptance

4.2.6.1 [TI-HOD-01] Test Cases and Procedures Overview

Test Item ID: TI-HOD-01

Test Case ID: TC-08-01

Test Case: HOD Access to Task Force Management

Priority: Very High

Test Technique: Use Case Testing, Positive testing

Test Scenario: HOD accesses Task Force Management.

SRS Traceability: SRS_REQ_301

Test Objective:

Verify that a user assigned the "Head of Department (HOD)" role can successfully access their department's task force management functions after logging in.

Prerequisites:

- A user account with the HOD role already exists.
- The HOD is assigned a valid department.
- The TFMS application is accessible.

Test Data:

- A valid HOD username and password.



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Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	14 /120

Test Procedure:

1. Log in to TFMS using HOD credentials.
2. Observe the dashboard or main menu displayed after login.
3. In the main menu or dashboard, view "department's task forces".
4. Click this link.

Expected Result:

- HOD successfully logs in.
- The HOD dashboard contains a clear entry point to its department's task group management.
- Clicking this will load the "Task Group Management" interface, which should display the task groups under its department.

Postconditions:

- HOD is located in its department's task group management interface.
- The displayed functionality and data scope should be limited to its department.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	15 /120

Test Case ID: TC-08-02

Test Case: TFMS Displays Department Task Groups

Priority: Very High

Testing Techniques: Use Case Testing, Positive testing

Test Scenario: TFMS displays department task forces.

SRS Traceability: SRS_REQ_302

Test Objective:

To verify that when a HOD accesses the task group management module, the system can correctly list all task groups belonging to the HOD's department.

Prerequisites:

- The HOD has successfully logged in and accessed the task group management module.
- At least one task group belonging to the HOD's department exists in the system.

Test Data:

- A list of known task group names belonging to this department.

Test Procedure:

1. The HOD logs in and enters the task group management page.
2. View the displayed task group list or panel.
3. Check each task group appearing in the list to confirm that its "Department" attribute matches the HOD's department.
4. Check if there are any task groups that do not belong to this department.

Expected Result:

- The task group list loads successfully.
- The list contains all task groups belonging to this HOD department.
- Task group information in the list includes: Name, Departments, Description, Status, and Weightage.
- Task groups from other departments should not appear.

Postcondition:



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- The HOD views the list view of its department task groups.

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SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	17 /120

Test Case ID: TC-08-03

Test Case: HOD filters and selects a task force

Priority: Very High

Testing Techniques: Use Case Testing, Positive testing

Test Scenario: HOD filters and selects a task force.

SRS Traceability: SRS_REQ_303

Test Objective:

To verify that the HOD can use the system's filtering function to narrow down the list of task forces and select a specific task force for viewing or management.

Prerequisites:

- The HOD is logged in and located on its department's task force management page.
- Multiple task forces with different statuses exist within this department.

Test Data:

- Filtering Criteria: Select "Status" as "Inactive".
- Target Selection: The name of a task force in the "Inactive" status.

Test Procedure:

1. Locate the filter control above the list on the task force management page.
2. In the "Status" filter, select "Inactive".
3. Observe the list refresh automatically after filtering.
4. Confirm that the list only displays task forces with the "Inactive" status.
5. Select a target "Inactive" task group from the filtered list and click its name or the "Manage" button.

Expected Result:

- After applying the "Inactive" filter, the list will only display task groups with matching statuses.
- After successfully selecting a task group, the system should navigate to the task group's details page or member management interface.



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- The page title or context information should clearly display the name of the selected task group.

Post-conditions:

- The HOD enters the management context of the selected task group and can perform operations such as viewing, adding, and removing members.

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SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	2 /120

Test Case ID: TC-08-04

Test Case: TFMS displays current members of a selected task force

Priority: Very High

Test Technique: Use Case Testing, Positive testing

Test Scenario: TFMS displays the current members of the selected task force.

SRS Traceability: SRS_REQ_304

Test Objective:

To verify that when the HOD selects a task force, the system can accurately display the current member list of that task force.

Prerequisites:

- The HOD is logged in and has selected a specific task force from the department's task force list.
- Several members have been pre-assigned to this task force.

Test Data:

- The name of a task force with known assigned members.
- A list of known member names and roles in this task force.

Test Procedure:

1. The HOD selects the target task force and enters its management page.
2. The "Current Members" section can be viewed on the page.
3. View the displayed member list.
4. Compare the information in the list with the known test data.

Expected Results:

- The system clearly displays a member list.
- The list includes the name of each member.
- The list clearly indicates the role each member plays in the task group.
- The displayed information is consistent with the preset test data.

Post-Conditions:

- HOD can clearly view the current personnel composition of the task group.



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SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	3 /120

Test Case ID: TC-08-05-01

Test Case: HOD adds member to task group and updates workload

Priority: Very High

Testing Techniques: Use Case Testing, Positive Testing

Test Scenario: HOD adds a member to the task force; TFMS updates the member list and recalculates workload (normal workload)

SRS Traceability: SRS_REQ_305

Test Objective:

Verify that HOD can add a new member to the task group from the list of eligible employees, and that the system can immediately update the member list and trigger a recalculation of the workload for the relevant employee.

Prerequisites:

- HOD has selected a task group and entered the member management interface.
- There are eligible employees in the department who are active and not currently assigned to this task group.
- The system's workload calculation function is working correctly.

Test Data:

- Target task group name.
- Name of the eligible employee to be added.

Test Procedure:

1. In the task group member management interface, find the "Add Member" section.
2. The system displays a drop-down list showing available employees.
3. Select Instructor Test1 from the list.
4. Click "SUBMIT FOR APPROVAL".
5. Observe the page:
 - Confirm that Instructor Test1 appears in the updated member list.
 - Check if the system displays any information about workload recalculation.



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SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	4 /120

Expected Results:

- The system confirms the "Member added successfully" message.
- The task group's member list updates in real time, including the newly added Instructor Test1.
- The system logs this change and triggers the workload recalculation logic.
- The total workload for affected employees is updated.

Postconditions:

- The task group member list has been updated.
- The workload data for the relevant employees has been recalculated.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	5 /120

Test Case ID: TC-08-05-02

Test Case: HOD adds member to task group and updates workload

Priority: Very High

Testing Techniques: Use Case Testing, Positive Testing

Test Scenario: HOD adds a member to the task force; TFMS updates the member list and recalculates workload (overloaded workload).

SRS Traceability: SRS_REQ_305

Test Objective:

Verify that when the HOD adds a new member to the task group, TFMS updates the member list and recalculates workload, resulting in an overloaded workload condition that will be referenced in subsequent warning and override scenarios.

Prerequisites:

- HOD has selected a task group and entered the member management interface.
- There are eligible employees in the department who are active and not currently assigned to this task group.
- The system's workload calculation function is working correctly.
- At least one employee is already near workload capacity.

Test Data:

- Target task group name.
- Name of the eligible employee to be added (Instructor Test1).

Test Procedure:

1. In the task group member management interface, find the "Add Member" section.
2. The system displays a drop-down list showing available employees.
3. Select Instructor Test1 from the list.
4. Click "SUBMIT FOR APPROVAL".
5. Observe the page:
 - Confirm that Instructor Test1 appears in the updated member list.



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SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
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- Check if the system displays any information about workload recalculation.

Expected Results:

- The system acknowledges the submission with a confirmation message.
- The task group's member list updates in real time, showing Instructor Test1 as pending approval.
- The system logs this change and triggers workload recalculation logic.
- The recalculation updates the total workload for affected employees, which may result in an overloaded workload state.
- No warning or override prompt is displayed at this stage, these are validated in subsequent test cases (TC-08-07, TC-08-10, TC-08-11).

Postconditions:

- The task group member list reflects the newly added member in a pending approval state.
- Workload data for the relevant employees has been recalculated.
- If the workload recalculation results in an overload, TFMS stores this state in the system logs and marks the affected employee(s) as overloaded. This recorded state will be explicitly checked in subsequent test cases (TC-08-07, TC-08-10, TC-08-11).



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SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	7 /120

Test Case ID: TC-08-05-03

Test Case: HOD removes member from task group and updates workload

Priority: Very High

Testing Techniques: Use Case Testing, Negative Testing

Test Scenario: HOD removes a member from the task force; TFMS updates the member list and recalculates workload.

SRS Traceability: SRS_REQ_305

Test Objective:

To verify that HOD can remove an existing member from a task group, and that the system can immediately update the member list and trigger a recalculation of the workload for the relevant employee.

Prerequisites:

- HOD has selected a task group and entered the member management interface.
- There must be at least one member in the task group who can be removed.
- The system's workload calculation function is working correctly.

Test Data:

- Target task group name.
- Existing member name to be removed: Test1.

Test Procedure:

1. Find member Test1 in the task group member list.
2. Click the "Delete" button next to the member's record.
3. When prompted for confirmation, click "Confirm Deletion".
4. Observe the following on the page:
 - Confirm that Test1 has disappeared from the updated member list.
 - Check if the system displays any information about workload recalculation.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	1 / 120

Expected Results:

- The system confirms the "Member Removal Successfully" message.
- The task group's member list is updated in real time and no longer contains Test1.
- The system records this change and triggers the workload recalculation logic.
- The removed employee's total workload is updated.

Postconditions:

- The task group member list has been updated.
- The workload data for the relevant employees has been recalculated.

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Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	2 /120

Test Case ID: TC-08-06

Test Case: TFMS calculates workload status for all task force members

Priority: Very High

Testing Techniques: Calculation Testing, Positive testing

Test Scenario: TFMS calculates workload status for all task force members.

SRS Traceability: SRS_REQ_306

Test Objective:

Verify that the system can automatically calculate and display the current total workload status for all members of a specified task force based on predefined thresholds.

Prerequisites:

- The HOD has selected a task force and entered the member management interface.
- The members of this task force have been assigned different roles.
- The system has configured workload classification thresholds.

Test Data:

- A task force containing multiple members.
- The allocation of each member in other task forces is known.

Test Procedure:

1. The HOD accesses the page displaying task force members and their workload status.
2. View the "Workload Status" displayed for each member.
3. Record the displayed status.



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Expected Results:

- The system displays a clear workload status classification for each member in a task group.
- The status classification is based on a comparison of the member's total workload (weighted sum) across all task groups with a system threshold.
- The displayed status is clearly indicated and easy to understand.

Postconditions:

- HOD provides a clear understanding of each member's workload.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	4 /120

Test Case ID: TC-08-07

Test Case: TFMS warns HOD when a member is overloaded

Priority: Very High

Testing Techniques: Use Case Testing, Negative Testing/Boundary Testing

Test Scenario: TFMS warns HOD if any member is overloaded.

SRS Traceability: SRS_REQ_307

Test Objective:

Verify that when an action performed by the HOD (such as adding a member) causes a member's total workload to exceed the "overload" threshold, the system can issue a clear warning message to the HOD.

Prerequisites:

- The HOD is logged in and ready to add a new member to a task group or change a member's role.
- There is an employee in the system whose current total workload is close to or has reached the upper limit of the "balanced" state.
- The system has correctly defined the "overload" threshold.

Test Data:

- Target task group name.
- Name of an employee nearing full load: Overloaded Employee A.
- Attempt to assign a new role (such as Chairman, which has higher weight) to Overloaded Employee A.



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Test Procedure:

1. The HOD accesses the member management interface of the target task group.
2. Perform an operation that may cause high-load employee A to become overloaded, such as:
 - Attempt to add high-load employee A to this task group and assign the chairperson role.
 - Attempt to change high-load employee A's role in this task group from member to chairperson.
3. Observe the system interface during the confirmation steps before/after performing the operation.

Expected Result:

- The system detects that this operation will cause high-load employee A to enter an "overloaded" state.
- The system displays a clear warning message to the HOD, mentioning the "overload" risk, the name of the affected employee, and the potential impact.
- Warning message appears.
- The system may ask the HOD to confirm whether to continue or suggest selecting another employee.

Post-conditions:

- The HOD receives the warning and can decide to continue or cancel the operation.
- If the HOD chooses to continue, the system should record this decision.

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SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	6 /120

Test Case ID: TC-08-08

Test Case: HOD submits task group for approval

Priority: Very High

Testing Techniques: Use Case Testing, Positive Testing

Test Scenario: HOD submits a task group for approval.

SRS Traceability: SRS_REQ_308

Test Objective:

Verify that HOD can successfully initiate the submission process for a completed task group member list, changing its status from "Draft" to "Submitted for Approval".

Prerequisites:

- HOD is logged in and has finalized the member list of a task group with a status of "Draft".
- The task group meets the minimum submission requirements.
- The submission function is available.

Test Data:

- The name of a task group in "Draft" status, ready for submission.

Test Procedure:

1. HOD navigates to the target task group's management page.
2. Find and click the "SUBMIT FOR APPROVAL" button.
3. Observe the system confirmation message and the task group's status indicator.

Expected Result:

- In the list view, the task group's status changes from "Draft" to "Submitted".
- The HOD's ability to modify members of this task group is restricted.
- The submission operation is logged.



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Postconditions:

- The task group list is locked, and the HOD cannot edit it further.
- The task group appears in the PSM's approval queue.

Test Case ID: TC-08-09

Test Case: TFMS sends submissions to PSM and logs them.

Priority: Very High

Testing Techniques: Integration Testing, Positive Testing

Test Scenario: TFMS sends submissions to PSM, and TFMS logs the submission operation.

SRS Traceability: SRS_REQ_309, SRS_REQ_310

Test Objective:

To verify that after an HOD submits a submission, the system automatically routes the task group list to the corresponding PSM (Project and Staff Manager) personnel and creates an audit log entry for that submission event.

Prerequisites:

- An HOD has just submitted a task group for approval (as described in TC-08-08).
- The PSM role exists and is assigned to at least one user in the system.
- The audit log function is enabled.

Test Data:

- The name of the most recently submitted task group.
- The username of the HOD who submitted the task group.

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Test Procedure:

1. After the HOD submits, log out of the HOD account.
2. Log in to TFMS using a user account with the PSM role.
3. Navigate to the PSM Approvals dashboard or the "Pending Submissions" section.
4. Check if the task group for the newly submitted HOD appears in the list.
5. Log Verification:
 - Log out of the PSM account.
 - Log in using an administrator account.
 - Navigate to the Audit Logs section.
 - Filter logs by HOD username and the type of operation related to "Submission".
 - Find the most recent log entry corresponding to the submission event.

Expected Results:

- Route to PSM: The submitted task group appears in the PSM Pending Approvals list and is accessible only to the PSM user.
- Audit Log Recording: The corresponding entry can be viewed in the audit logs.

Post-Conditions:

- The submission was successfully transferred to the PSM workflow.
- A permanent record of the submission operation exists in the audit logs.

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Test Case ID: TC-08-10

Test Case: HOD overload warning coverage

Priority: Very High

Testing Techniques: Use Case Testing, Decision Testing

Test Scenario: HOD Overload Warning Coverage.

SRS Traceability: SRS_REQ_311

Test Objective:

To verify that when the system displays an overload warning, HOD has an option to cover the warning and continue operation, while identifying the associated risks.

Prerequisites:

- HOD is performing an operation that triggers an overload warning (as described in TC-08-07).
- The system is configured to allow coverage and has appropriate accountability mechanisms.

Test Data:

- The same scenario as TC-08-07, i.e., attempting to add a high-load employee A as a task group chair triggers a warning.

Test Procedure:

- Trigger an overload warning by attempting to assign a high-load employee A as a task group chair.
- When the warning message appears, look for the "Confirm and Ignore Warning" option.
- Click the coverage option.
- The system requires a reason; please enter a brief explanation.
- Complete the operation.



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Expected Result:

- The system displays an overload warning, but also provides an explicit overwrite mechanism for the HOD.
- After overwrite, the system continues to perform the originally requested operation (e.g., a member is added).
- The system logs the overwrite operation and any provided reasons in the audit log or assignment history.

Postconditions:

- The operation was completed despite the initial warning.
- The audit trail records the overwrite operation and its reason.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	11 /120

Test Case ID: TC-08-11

Test Case: TFMS prompts HOD to provide a reason for coverage operation

Priority: Very High

Testing Techniques: Use Case Testing, Compliance Testing

Test Scenario: TFMS prompts HOD to provide a reason for coverage operation.

SRS Traceability: SRS_REQ_312

Test Objective:

Verify that when HOD selects to cover an overload warning, the system forces HOD to enter a text reason before allowing the operation to complete.

Prerequisites:

- HOD is performing an operation that triggers an overload warning and has selected to cover it (as described in steps 1-3 of TC-08-11).
- System policy requires a reason for coverage operation.

Test Data:

- Same scenario as TC-08-11.
- Reason text.

Test Procedure:

1. Trigger the overload warning and click the cover option.
2. Close Observation: After clicking cover, the system should not immediately complete the operation. Instead, a new input field or dialog box should be displayed, requiring a reason.
3. Try continuing the operation without entering any text in the reason field. Click "Commit".
4. Observe whether the system blocks the commit.
5. Enter the reason text into the provided field.
6. Click "Commit" again.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	12 /120

Expected Result:

- After initiating the overwrite, the system prompts the HOD to fill in the reason field.
- If the HOD attempts to continue with an empty reason, the system will display a validation error (e.g., "A reason must be provided") and will not complete the operation.
- The system will only complete the operation and record the overwrite operation and the provided reason after a non-empty reason is provided.

Postcondition:

- Operation completed.
- A non-empty reason is permanently stored along with the transaction record.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	13 /120

The requirements in the SRS_REQ_400, which focus on approval workflow, are addressed through Test Model TM-PSM-01. This model refers to TI-PSM-01, Task Force Approval by PSM, covering SRS_REQ_401–409, ensuring approval processes and workflow integrity are tested.

4.2.7 Test Item: Task Force Approval (PSM)

Test Item ID: TI-PSM-01

Test Item Name: Task Force Approval (PSM)

Priority: Very High

Test Level: System / Acceptance

4.2.7.1 [TI-PSM-01] Test Cases and Procedures Overview

Test Item ID: TI-APPROVAL-01

Test Case ID: TC-09-01

Test Case: Accessing the Approval Panel

Priority: Very High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: PSM access to the approval panel for reviewing task force submissions.

SRS Traceability: SRS_REQ_401

Test Objective:

Verify that the PSM user can successfully access the Approval Panel from the TFMS dashboard.

Preconditions:

- The PSM is logged into TFMS.
- The PSM role has appropriate permissions to access the approval panel.

Test Data:

- A valid PSM username and password.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	14 /120

Test Procedure:

1. The PSM logs in to TFMS using valid credentials.
2. After logging in, the PSM is taken to the dashboard interface.
3. Locate and click "VIEW APPROVALS" in the dashboard.
4. Observe the page loading and the displayed interface.

Expected Results:

- The approval panel page loads successfully.
- The interface displays a list of submitted task force assignments.

Postconditions:

- The PSM remains on the approval panel interface.
- No changes have been made to any submissions.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	15 /120

Test Case ID: TC-09-02

Test Case: Displaying submitted task force assignments

Priority: Very High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: TFMS displays all submitted task force assignments in the approval panel.

SRS Traceability: SRS_REQ_402

Test Objective:

Verify that TFMS correctly displays all pending task force submissions for PSM review.

Preconditions:

- The PSM is logged into TFMS and on the approval panel.
- At least one task force submission exists from HOD.

Test Data:

- Existing task force submissions from HOD.

Test Procedure:

1. Access the approval panel.
2. Observe the list of displayed submissions.
3. Verify that each submission shows relevant details (e.g., name, department, description, workload, role).

Expected Results:

- All pending submissions are displayed in a structured list.
- Each entry includes sufficient information for review.

Postconditions:

- Submissions remain unchanged and in pending status.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	16 /120

Test Case ID: TC-09-03

Test Case: PSM reviews and adjusts task force assignments

Priority: Very High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: PSM reviews submitted assignments and makes adjustments if needed.

SRS Traceability: SRS_REQ_403

Test Objective:

Verify that the PSM can review task force assignments and make adjustments to member assignments.

Preconditions:

- The PSM is on the approval panel with at least one submission.

Test Data:

- A pending task force submission with at least one assigned member.

Test Procedure:

1. Select a submission from the list.
2. Review the current assigned members.
3. Use the interface to add or remove members as needed.
4. Save the adjustments.

Expected Results:

- Adjustments are saved successfully.
- The updated member list is reflected in the submission preview.

Postconditions:

- The submission is updated with adjusted assignments.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	17 /120

Test Case ID: TC-09-04

Test Case: TFMS recalculates workload after adjustments

Priority: Very High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: TFMS recalculates workload for all members after PSM makes adjustments.

SRS Traceability: SRS_REQ_404

Test Objective:

Verify that TFMS automatically recalculates and updates workload status for all task force members after any assignment adjustment by PSM.

Preconditions:

- The PSM has just adjusted a task force assignment.

Test Data:

- Adjusted task force submission with updated member list.

Test Procedure:

1. After saving adjustments, observe the workload status display.
2. Verify that workload classifications (Balanced/OverLoaded) are updated.

Expected Results:

- Workload status is recalculated and displayed correctly.
- Overload warnings appear if any member exceeds thresholds.

Postconditions:

- Workload data is updated in the system.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	18 /120

Test Case ID: TC-09-05-01

Test Case: PSM approves a submission

Priority: Very High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: PSM approves a task force submission after review.

SRS Traceability: SRS_REQ_405

Test Objective:

Verify that the PSM can approve a task force submission and that TFMS locks the approved assignments.

Preconditions:

- The PSM has reviewed a submission and is ready to approve.

Test Data:

- A pending task force submission.

Test Procedure:

1. Select the submission.
2. Click the "APPROVE" button.
3. Confirm the approval action.
4. Click the "APPROVE TASK FORCE" button
5. Observe the system response.

Expected Results:

- The system confirms "Task Force ' ' approved."
- The assignment is locked and cannot be edited further.

Postconditions:

- The submission status changes to "Approved".
- Assignments are locked per R2: Approval Lock.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	19 /120

Test Case ID: TC-09-05-02

Test Case: PSM rejects a submission

Priority: Very High

Test Technique: Use Case Testing, Negative Testing

Test Scenario: PSM rejects a task force submission and provides feedback.

SRS Traceability: SRS_REQ_405, SRS_REQ_408

Test Objective:

Verify that the PSM can reject a submission and that TFMS prompts for and logs feedback.

Preconditions:

- The PSM has reviewed a submission and decides to reject.

Test Data:

- A pending task force submission.
- Rejection feedback text.

Test Procedure:

1. Select the submission.
2. Click the "REJECT SUBMISSION" button.
3. Enter feedback in the provided text field.
4. Confirm the rejection.

Expected Results:

- The system confirms "Task Force ' ' rejected."
- Feedback is saved and logged.
- Submission is returned to HOD for revision.

Postconditions:

- Submission status changes to "Rejected".
- Feedback is recorded in audit log.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	20 /120

Test Case ID: TC-09-06

Test Case: TFMS locks approved assignments

Priority: Very High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: TFMS automatically locks assignments once approved by PSM.

SRS Traceability: SRS_REQ_406

Test Objective:

Verify that approved task force assignments are locked and cannot be modified without administrative override.

Preconditions:

- A task force submission has been approved by PSM.

Test Data:

- An approved task force assignment.

Test Procedure:

1. Navigate to the task force management module as HOD.
2. Attempt to edit members in the approved assignment.
3. Observe system behavior.

Expected Results:

- Edit functions are disabled or prevented.
- System displays message: "This task force is locked while submitted, approved, or inactive."

Postconditions:

- Assignment remains unchanged and locked.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	21 /120

Test Case ID: TC-09-07

Test Case: TFMS logs all PSM actions

Priority: Very High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: TFMS records all actions performed by PSM in the audit log.

SRS Traceability: SRS_REQ_407

Test Objective:

Verify that every action taken by PSM in the approval panel is logged in the audit trail.

Preconditions:

- The PSM has performed at least one action (approve/reject/adjust).
- Audit logging is enabled.

Test Data:

- PSM user session.
- Action details (e.g., "UPDATE_TASKFORCE").

Test Procedure:

1. Perform an action in the approval panel.
2. Navigate to the audit log section as an administrator.
3. Filter logs by PSM user and date.
4. Verify the logged entry.

Expected Results:

- The action is recorded with timestamp, actor, action, target, details and IP address.
- Log entry is accurate and complete.

Postconditions:

- Audit log contains the new entry.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	22 /120

Test Case ID: TC-09-08

Test Case: PSM rejects a submission (negative flow)

Priority: Very High

Test Technique: Use Case Testing, Negative Testing

Test Scenario: PSM attempts to reject a submission without providing feedback.

SRS Traceability: SRS_REQ_408

Test Objective:

Verify that TFMS prevents rejection without mandatory feedback entry.

Preconditions:

- The PSM is reviewing a submission.

Test Data:

- A pending task force submission.

Test Procedure:

1. Click the "REJECT SUBMISSION" button.
2. Leave the feedback field blank.
3. Attempt to confirm rejection.

Expected Results:

- System prevents rejection and displays: "Rejection reason is required."
- Submission remains in pending status.

Postconditions:

- No changes to submission status.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	23 /120

Test Case ID: TC-09-09

Test Case: PSM enters feedback for rejected submission

Priority: Very High

Test Technique: Use Case Testing, Negative Testing

Test Scenario: PSM provides detailed feedback when rejecting a submission.

SRS Traceability: SRS_REQ_409

Test Objective:

Verify that PSM can enter and submit feedback for a rejected task force submission.

Preconditions:

- The PSM has selected to reject a submission.

Test Data:

- Rejection feedback text: "Adjust workload distribution before resubmission."

Test Procedure:

1. Enter feedback in the provided text area.
2. Click "REJECT SUBMISSION".
3. Verify submission status and feedback visibility.

Expected Results:

- Feedback is saved and displayed in the submission history.
- Submission status updates to "Rejected".

Postconditions:

- Feedback is attached to the submission log.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	24 /120

The requirements in the SRS_REQ_500, which cover lecturer portfolio management, are addressed through Test Model TM-LECT-01. This model refers to TI-LECT-01, Lecturer Portfolio, covering SRS_REQ_501–509, ensuring lecturer data and academic records are validated.

4.2.8 Test Item: Lecturer Portfolio

Test Item ID: TI-LECT-01

Test Item Name: Lecturer Portfolio

Priority: Very High

Test Level: System / Acceptance

4.2.8.1 [TI-LECT-01] Test Cases and Procedures Overview

Test Item ID: TI-LEC-01

Test Case ID: TC-10-01

Test Case: Lecturer accesses "My Portfolio" section

Priority: High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: Lecturer access to the personal task force portfolio.

SRS Traceability: SRS_REQ_501

Test Objective:

Verify that the lecturer user can successfully access the "My Portfolio" section from the TFMS dashboard.

Preconditions:

- The lecturer is logged into TFMS.
- The lecturer role has appropriate permissions to access personal portfolio.

Test Data:

- A valid lecturer username and password.

Test Procedure:



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	25 /120

1. The lecturer logs in to TFMS using valid credentials.
2. After logging in, navigate to the dashboard.
3. Locate and click "My Portfolio" in the menu.
4. Observe the page loading and the displayed interface.

Expected Results:

- The "My Portfolio" page loads successfully.
- The interface displays a dashboard showing task force assignments.

Postconditions:

- The lecturer remains on the portfolio interface.
- No changes have been made to any assignment data.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	26 /120

Test Case ID: TC-10-02

Test Case: TFMS displays dashboard with all task force assignments

Priority: High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: TFMS displays a personalized dashboard showing all current and historical task force assignments for the lecturer.

SRS Traceability: SRS_REQ_502

Test Objective:

Verify that TFMS correctly displays all task force assignments associated with the logged-in lecturer.

Preconditions:

- The lecturer is logged into TFMS and on the "My Portfolio" page.
- The lecturer has at least one approved task force assignment.

Test Data:

- Lecturer's assigned task force records.

Test Procedure:

1. Access "My Portfolio".
2. Observe the dashboard layout and content.
3. Verify that current assignments are listed with status.

Expected Results:

- Dashboard shows all current task force assignments.
- Each assignment includes departments, weightage, HOD, PSM, and members.

Postconditions:

- Assignment data remains unchanged.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	27 /120

Test Case ID: TC-10-03

Test Case: Lecturer views task force history

Priority: High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: Lecturer accesses historical records of past task force assignments.

SRS Traceability: SRS_REQ_503

Test Objective:

Verify that the lecturer can view historical task force assignments from previous periods.

Preconditions:

- The lecturer is on the "My Portfolio" page.
- Historical assignment records exist in the system.

Test Data:

- Lecturer's past task force assignments.

Test Procedure:

1. Locate and click the "History" or "Past Assignments" tab.
2. Observe the list of historical assignments.
3. Verify that details such as task force name, period, and role are displayed.

Expected Results:

- Historical assignments are displayed in a structured list.
- Each entry includes relevant historical data.

Postconditions:

- No changes to historical data.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	28 /120

Test Case ID: TC-10-04-01

Test Case: Lecturer downloads summary successfully (PDF)

Priority: High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: Lecturer successfully downloads a PDF summary of task force assignments.

SRS Traceability: SRS_REQ_504

Test Objective:

Verify that the lecturer can generate and download a PDF summary of their task force assignments.

Preconditions:

- The lecturer is on the DASHBOARD page.
- At least one assignment exists.

Test Data:

- Current task force assignments.

Test Procedure:

1. Click the "Download REPORT" button.
2. Select PDF as the format.
3. Confirm the download action.
4. Verify the downloaded file.

Expected Results:

- System generates a PDF file successfully.
- File contains accurate assignment details.
- Download completes without error.

Postconditions:

- PDF file is saved locally.
- No data is altered in TFMS.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	29 /120

Test Case ID: TC-10-04-02

Test Case: PDF generation fails; system displays error

Priority: High

Test Technique: Use Case Testing, Negative Testing

Test Scenario: System fails to generate PDF and displays an appropriate error message.

SRS Traceability: SRS_REQ_504, SRS_REQ_508

Test Objective:

Verify that TFMS handles PDF generation failures gracefully and informs the user.

Preconditions:

- The lecturer is on the "DASHBOARD" page.
- System is configured to simulate PDF generation failure.

Test Data:

- Current task force assignments.

Test Procedure:

1. Click the "DOWNLOAD REPORT" button.
2. Select PDF format.
3. Observe system response.

Expected Results:

- System displays an error message: "PDF generation failed. Please try again later."
- No file is downloaded.

Postconditions:

- System remains in the portfolio view.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	30 /120

Test Case ID: TC-10-05

Test Case: Lecturer submits remarks for task forces

Priority: High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: Lecturer submits remarks or feedback on a specific task force assignment.

SRS Traceability: SRS_REQ_505

Test Objective:

Verify that the lecturer can submit remarks for a task force assignment and that the system records and notifies appropriately.

Preconditions:

- The lecturer is on the "My Portfolio" page.
- At least one assignment is available for remark submission.

Test Data:

- Remark text: "Request for workload review."

Test Procedure:

1. Select a task force assignment.
2. Click "SEND REMARKS".
3. Enter remarks in the provided text field.
4. Click "SEND REMARKS".
5. Observe confirmation message.

Expected Results:

- System confirms "Your remarks have been sent to the PSM."
- Remarks are saved and attached to the assignment.

Postconditions:

- Remarks are logged and notification is sent to HOD/PSM per SRS_REQ_506.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	31 /120

Test Case ID: TC-10-06

Test Case: TFMS notifies HOD/PSM after remarks submission

Priority: High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: System sends a notification to HOD and/or PSM when a lecturer submits remarks.

SRS Traceability: SRS_REQ_506

Test Objective:

Verify that TFMS automatically notifies the relevant authorities (HOD/PSM) when a lecturer submits remarks.

Preconditions:

- Lecturer has just submitted a remark.
- Notification system is operational.

Test Data:

- Submitted remark details.
- HOD/PSM contact information.

Test Procedure:

1. Submit remarks as lecturer.
2. Check HOD/PSM notification inbox.
3. Verify notification content.

Expected Results:

- HOD/PSM receive a notification indicating new remarks from the lecturer.
- Notification includes assignment details and remark text.

Postconditions:

- Notification is logged in the system.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	32 /120

Test Case ID: TC-10-07

Test Case: Lecturer accesses portfolio but has no assignments

Priority: High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: Lecturer accesses "My Portfolio" but has no task force assignments in the current session.

SRS Traceability: SRS_REQ_507, SRS_REQ_508

Test Objective:

Verify that TFMS displays an appropriate message when the lecturer has no current task force assignments.

Preconditions:

- The lecturer is logged into TFMS.
- The lecturer has no active task force assignments.

Test Data:

- Lecturer with zero current assignments.

Test Procedure:

1. Access "My Portfolio".
2. Observe the displayed message.

Expected Results:

- System displays: "You have no task force assignments."
- Dashboard shows no assignment entries.

Postconditions:

- No changes to system data.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	33 /120

Test Case ID: TC-10-08

Test Case: Lecturer Portfolio – PDF download failure handling

Priority: High

Test Technique: Use Case Testing, Negative Testing, Error Handling

Test Scenario: Lecturer attempts to download portfolio summary as PDF, but the system encounters an error during PDF generation or download.

SRS Traceability: SRS_REQ_509

Test Objective:

To verify that TFMS correctly handles PDF generation or download failures by displaying an appropriate error notification without affecting system data..

Preconditions:

- Lecturer is successfully logged into TFMS.
- Lecturer has at least one task force assignment available in "My Portfolio".
- System is configured to simulate or trigger an error during PDF generation/download (e.g., service unavailable, timeout, or file generation failure).

Test Data:

- Lecturer account with valid portfolio data.
- Simulated PDF generation error condition.

Test Procedure:

1. Lecturer accesses the "My Portfolio" section.
2. Lecturer clicks "Download Summary (PDF)".
3. TFMS attempts to generate and download the PDF file.
4. Observe system response when PDF generation/download fails.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	34 /120

Expected Results:

- PDF file is not downloaded.
- TFMS displays a clear error notification (e.g., "Failed to generate PDF. Please try again later.").
- System remains responsive and does not crash.

Postconditions:

- No changes are made to lecturer portfolio data.
- No incomplete or corrupted PDF file is stored or downloaded.

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	35 /120

The requirements in the SRS_REQ_600, which emphasize executive reporting and audit functions, are addressed through Test Models TM-MGMT-01. TI-MGMT-01, Executive Reports) covers SRS_REQ_601–606. These items ensure reporting accuracy.

4.2.9 Test Item: Executive Reports

Test Item ID: TI-MGMT-01

Test Item Name: Executive Reports

Priority: Very High

Test Level: System / Acceptance

4.2.9.1 [TI-MGMT-01] Test Cases and Procedures Overview

Test Item ID: TI-MGMT-01

Test Case ID: TC-11-01

Test Case: Management accesses Executive Reports section

Priority: High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: Faculty Management access to the Executive Reports and Analytics module.

SRS Traceability: SRS_REQ_601

Test Objective:

Verify that the Management user can successfully access the Executive Reports section from the TFMS dashboard.

Preconditions:

- Management user is logged into TFMS.
- Management role has appropriate permissions to access executive reports.

Test Data:

- A valid management username and password.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	36 /120

Test Procedure:

- 1. Log in to TFMS as Management.
- 2. Navigate to the dashboard.
- 3. Locate and click "Audit Logs" in the menu.
- 4. Observe page loading and interface.

Expected Results:

- The Executive Reports page loads successfully.
- Interface displays a high-level dashboard with key metrics.

Postconditions:

- Management remains on the executive reports interface.

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DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	37 /120

Test Case ID: TC-11-02

Test Case: TFMS displays Executive Reports dashboard

Priority: High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: TFMS displays an executive dashboard with timestamp, actor, action, details, and IP address.

SRS Traceability: SRS_REQ_602

Test Objective:

Verify that the executive dashboard correctly displays aggregated data.

Preconditions:

- Management is logged into TFMS and on the Executive Reports page.
- Sufficient data exists for reporting.

Test Data:

- Aggregated metrics data.

Test Procedure:

1. Access Executive Reports.
2. Review the dashboard layout and content.
3. Verify that key metrics are displayed clearly.

Expected Results:

- Dashboard shows timestamp, actor, action, details, and IP address.
- Data is presented in tables.

Postconditions:

- Dashboard data is read-only.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	38 /120

Test Case ID: TC-11-03

Test Case: Management drills down to department-level details

Priority: High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: Management drills down from the executive dashboard to view department-specific details.

SRS Traceability: SRS_REQ_603

Test Objective:

Verify that management can navigate from the high-level dashboard to detailed department-level views.

Preconditions:

- Management is on the Executive Reports dashboard.
- Department data is available.

Test Data:

- Department identifiers (e.g., "Computer Science", "Engineering").

Test Procedure:

1. Click "MANAGE DEPARTMENTS" button.
2. Observe the department-level view.
3. Verify that details such as department name, staff count, and task forces are shown.

Expected Results:

- System transitions to a department detail view.
- View includes department name, staff count, and task forces.

Postconditions:

- Management is viewing department-level data.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	39 /120

Test Case ID: TC-11-04-01

Test Case: Management generates a report: All Task Force

Priority: High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: Management configures and generates a custom report using filters and parameters.

SRS Traceability: SRS_REQ_604

Test Objective:

Verify whether management can generate reports: All Task Force.

Preconditions:

- Management is on the Management Page.
- Report generation functionality is available.

Test Data:

- Track system security and user actions.

Test Procedure:

1. Click "EXPORT TO CSV".
2. Observe system responses.

Expected Results:

- System processes the request.
- Report generates.

Postconditions:

- All Task Force Report has been exported in csv format.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	40 /120

Test Case ID: TC-11-04-02

Test Case: Management generates a report: Department-wise

Priority: High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: Management configures and generates a custom report using filters and parameters.

SRS Traceability: SRS_REQ_604

Test Objective:

Verify whether management can generate reports: Department-wise

Preconditions:

- Management is on the Management Page.
- Report generation functionality is available.

Test Data:

- Track system security and user actions.

Test Procedure:

3. Click "EXPORT TO CSV".
4. Observe system responses.

Expected Results:

- System processes the request.
- Report generates.

Postconditions:

- Department-wise Report has been exported in csv format.



DOCUMENT IDENTIFICATION

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Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	41 /120

Test Case ID: TC-11-05

Test Case: TFMS exports report

Priority: High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: Management exports a generated report.

SRS Traceability: SRS_REQ_605

Test Objective:

Verify that TFMS can export a report and provide a downloadable file.

Preconditions:

- A report has been generated and is ready for export.

Test Data:

- Generated report data.

Test Procedure:

1. Click "EXPORT TO CSV".
2. Verify file download.

Expected Results:

- System generates and downloads a file.
- File contains accurate report data.

Postconditions:

- Report file is saved locally.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	42 /120

Test Case ID: TC-11-06

Test Case: TFMS logs all activities related to report export

Priority: High

Test Technique: Use Case Testing, Positive Testing

Test Scenario: System logs all actions performed by management in the executive reports module.

SRS Traceability: SRS_REQ_606

Test Objective:

Verify that TFMS records all report-related activities (export) in the audit log.

Preconditions:

- Management has performed at least one action in the executive report module.
- Audit logging is enabled.

Test Data:

- Management user session details.
- Action descriptions (e.g., "Generated department report for Engineering").

Test Procedure:

1. Perform an action in the executive report's module.
2. Navigate to audit logs as an administrator.
3. Filter logs by management user and module.
4. Verify the logged entry.

Expected Results:

- Each action is logged with timestamp, actor, action, and details.
- Log entries are accurate and complete.

Postconditions:

- Audit log is updated with new entries.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	43 /120

4.2.10 Test Item: Audit and Notification

Test Item ID: TI-AUD-01

Test Item Name: Audit and Notification

Priority: Very High

Test Level: System / Acceptance

4.2.10.1 [TI-AUD-01] Test Cases and Procedures Overview

Test Case ID: TC-06-01

Test Case: Administrator viewing audit logs

Priority: High

Testing Techniques: Use Case Testing, Positive testing

Test Scenario: Administrator viewing audit logs.

SRS Traceability: SRS_REQ_211

Test Objective:

Verify that the administrator can successfully access the audit log module and view the historical list of system activities.

Prerequisites:

- The administrator is logged into TFMS.
- Some historical activity records exist in the system.
- The "Audit Logs" module is accessible.

Test Data:

- No specific test data; relies on existing system logs.

Test Procedure:

1. Navigate to the "Audit Logs" module.
2. Observe the log list after the page loads.



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3. Check if the list contains the following typical columns: TIMESTAMP, ACTOR, ACTION, TARGET, DETAILS, and IP ADDRESS.
4. Verify if the list contains some recent activity records.

Expected Result:

- The audit log page loads successfully.
- The log list is clearly displayed and contains multiple records.
- The list includes relevant column information for easy reading.
- The list contains records of recent actions performed by the administrator (e.g., "Add Employee").

Postconditions:

- The administrator is on the audit log interface.
- No modifications have been made to the log content.

**DOCUMENT IDENTIFICATION**

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Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	45 /120

Test Case ID: TC-06-02

Test Case: TFMS logs all administrator operations

Priority: High

Testing Techniques: Audit Testing, Positive testing

Test Scenario: TFMS logs all actions performed by the Admin.

SRS Traceability: SRS_REQ_212

Test Objective:

To verify that the system can automatically record all key operations performed by the administrator in TFMS, and that these records can be queried in the audit logs.

Prerequisites:

- The administrator is logged into TFMS.
- The audit log function is enabled.
- The administrator is preparing to perform some loggable operations.

Test Data:

- Operation to be performed: Add a new test employee.

Test Procedure:

1. Perform a clear administrative operation: Navigate to Employee Management, add a new employee, and save.
2. Navigate to the "Audit Logs" module.
3. Use the filter function (search by username) to find relevant records.
4. Find the log entry corresponding to step 1 in the results list.



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Expected Result:

- A record should be found in the audit log, with the following details:
 - TIMESTAMP: The time the operation occurred
 - ACTOR: The administrator username that performed the operation
 - ACTION: "CREATE_USER"
 - TARGET: User #6
 - DETAILS: Created user Test1
 - IP ADDRESS: 175.143.63.111

Postconditions:

- The administrator operation was successfully logged in the audit log.
- The log entry cannot be modified or deleted by the administrator or any user.

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5 NOTES

Acronyms Used:

- **CDRL** Context Data Requirement List
- **CSC** Computer Software Component
- **CSCI** Computer Software Configuration Item
- **CSU** Computer Software Unit
- **DOD** Department of Defense
- **FAI** Faculty of Artificial Intelligence
- **MIL** Military
- **OOAD** Object Oriented Analysis Design
- **SDD** Software Design Document
- **SDP** Software Development Plan
- **SRS** Software Requirements Specification
- **STD** Software Test Description
- **STR** Software Test Report
- **TC** Test Case
- **TFMS** Taskforce Management System
- **UML** Unified Modeling Language
- **URS** User Requirement Specification
- **UTM** Universiti Teknologi Malaysia

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	SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
	Task Force Management System (TFMS)	STD-TFMS-190126-V1-	A4	1.0	48 /120

6 ANNEX 1: REQUIREMENT TRACEABILITY MATRIX

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DOCUMENT IDENTIFICATION

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ANNEX 1: Requirement Traceability Matrix - Task Force Management System (TFMS)

No.	Test Model	Test Item	SRS IDs	Test Case ID	Test Scenario / Description	Priority
1.	Authentication and Access Control	User Authentication & Login	SRS_REQ_101,102	TC-01-01	User navigates to login page; TFMS displays login form	Very High
2.			SRS_REQ_103, 104, 105	TC-01-02	User enters valid username and password; input accepted, TFMS validates credentials: success	Very High
3.			SRS_REQ_103, 104, 105	TC-01-03-01	User enters invalid credentials; credentials: fail (invalid username and password)	Very High
4.			SRS_REQ_103, 104, 105	TC-01-03-02	User enters invalid credentials; credentials: fail (invalid username)	Very High
5.			SRS_REQ_103, 104, 105	TC-01-03-03	User enters invalid credentials; credentials: fail (invalid password)	Very High
6.			SRS_REQ_103, 104, 105	TC-01-04-01	User enters empty username and password; displays mandatory field warnings; credentials: fail	Very High
7.			SRS_REQ_103, 104, 105	TC-01-04-02	User enters empty username; displays mandatory field warnings; credentials: fail	Very High
8.			SRS_REQ_103, 104, 105	TC-01-04-03	User enters empty password; displays mandatory field warnings; credentials: fail	Very High
9.			SRS_REQ_106	TC-01-05	TFMS checks if account is locked; displays lock message if locked	Very High
10.			SRS_REQ_107	TC-01-06	TFMS checks if first login; prompts for password change if required	Very High
11.			SRS_REQ_108	TC-01-07	TFMS creates session for user after successful login	Very High
12.			SRS_REQ_109	TC-01-08	TFMS displays role-appropriate dashboard	Very High
13.			SRS_REQ_115,116,117, 118,119	TC-01-09	Alternative/negative flow: Account locked after multiple failed login attempts	Very High
14.		Password Reset	SRS_REQ_110,111	TC-02-01	User clicks "Forgot Password?" and TFMS sends reset link to email	Very High

ANNEX 1: Requirement Traceability Matrix - Task Force Management System (TFMS)

No.	Test Model	Test Item	SRS IDs	Test Case ID	Test Scenario / Description	Priority
15.	Staff and Task Force Management		SRS_REQ_112,113	TC-02-02	TFMS detects unmodified initial password; displays password change form	Very High
16.			SRS_REQ_114	TC-02-03	User enters new password; system updates password successfully	Very High
17.		Staff Management	SRS_REQ_201	TC-03-01	Admin accesses Staff Management	Very High
18.			SRS_REQ_202, 203	TC-03-02-01	Admin adds staff records, TFMS auto-generates Staff IDs upon staff creation	Very High
19.			SRS_REQ_202, 203	TC-03-02-02	Admin edits staff records	Very High
20.			SRS_REQ_204	TC-03-03	Admin assigns user roles to staff	High
21.			SRS_REQ_205	TC-03-04	TFMS creates accounts with temporary passwords	Very High
22.			SRS_REQ_206	TC-03-05	Admin resets passwords and unlocks accounts	Very High
23.			SRS_REQ_207	TC-03-06	Admin deactivates staff members	High
24.		Task Force Management	SRS_REQ_208, 209	TC-04-01	Admin creates task forces, TFMS auto-generates Task Force IDs	Very High
25.			SRS_REQ_213	TC-07-01	Admin deactivates staff members (alternative flow)	High
26.			SRS_REQ_214	TC-07-02	Admin deactivates task forces	High
27.		System Configuration	SRS_REQ_210	TC-05-01	Admin configures TFMS parameters	High
28.	Audit and Notification	Audit Logging	SRS_REQ_211	TC-06-01	Admin views audit logs	High
29.			SRS_REQ_212	TC-06-02	TFMS logs all actions performed by Admin	High
30.			SRS_REQ_407	TC-09-07	TFMS logs all actions performed by PSM	Very High
31.			SRS_REQ_606	TC-11-06	TFMS logs all activities related to report access and export	High
32.	Assignment and Workload Management	Task Force Assignment (HOD)	SRS_REQ_301	TC-08-01	HOD accesses Task Force Management	Very High
33.			SRS_REQ_302	TC-08-02	TFMS displays department task forces	Very High
34.			SRS_REQ_303	TC-08-03	HOD filters and selects a task force	Very High
35.			SRS_REQ_304	TC-08-04	TFMS displays current members of the selected task force	Very High

ANNEX 1: Requirement Traceability Matrix - Task Force Management System (TFMS)

No.	Test Model	Test Item	SRS IDs	Test Case ID	Test Scenario / Description	Priority
36.			SRS_REQ_305	TC-08-05-01	HOD adds a member to the task force; TFMS updates member list and recalculates workload (normal workload)	Very High
37.			SRS_REQ_305	TC-08-05-02	HOD adds a member to the task force; TFMS updates member list and recalculates workload (overloaded workload)	Very High
38.			SRS_REQ_305	TC-08-05-03	HOD removes a member from the task force; TFMS updates member list and recalculates workload	Very High
39.			SRS_REQ_306	TC-08-06	TFMS calculates workload status for all task force members	Very High
40.			SRS_REQ_307	TC-08-07	TFMS warns HOD if any member is overloaded	Very High
41.			SRS_REQ_308	TC-08-08	HOD submits task force for approval	Very High
42.			SRS_REQ_309, 310	TC-08-09	TFMS sends submission to PSM, TFMS logs submission actions	Very High
43.			SRS_REQ_311	TC-08-10	HOD overrides overload warning	Very High
44.			SRS_REQ_312	TC-08-11	TFMS prompts HOD to provide justification for override	Very High
45.	Approval Workflow	Task Force Approval (PSM)	SRS_REQ_401	TC-09-01	PSM accesses Approval Panel	Very High
46.			SRS_REQ_402	TC-09-02	TFMS displays all task force submissions	Very High
47.			SRS_REQ_403	TC-09-03	PSM reviews and adjusts task force assignments	Very High
48.			SRS_REQ_404	TC-09-04	TFMS recalculates workload after PSM adjustments	Very High
49.			SRS_REQ_405	TC-09-05-01	PSM approves the submission; TFMS locks approved assignments	Very High
50.			SRS_REQ_405	TC-09-05-02	PSM rejects the submission; TFMS requires feedback entry	Very High
51.			SRS_REQ_406	TC-09-06	TFMS locks approved assignments independently of approval action	Very High
52.			SRS_REQ_407	TC-09-07	TFMS logs all actions performed by PSM	Very High
53.			SRS_REQ_408	TC-09-08	PSM rejects a submission	Very High

ANNEX 1: Requirement Traceability Matrix - Task Force Management System (TFMS)

No.	Test Model	Test Item	SRS IDs	Test Case ID	Test Scenario / Description	Priority
54.			SRS_REQ_409	TC-09-09	PSM enters feedback for rejected submissions	Very High
55.	Lecturer Portfolio	Lecturer Portfolio	SRS_REQ_501	TC-10-01	Lecturer accesses "My Portfolio" section	High
56.			SRS_REQ_502	TC-10-02	TFMS displays dashboard showing all task force assignments	High
57.			SRS_REQ_503	TC-10-03	Lecturer views task force history	High
58.			SRS_REQ_504	TC-10-04-01	Lecturer downloads summary successfully (PDF)	High
59.			SRS_REQ_504	TC-10-04-02	PDF generation fails; system displays error message	High
60.			SRS_REQ_505	TC-10-05	Lecturer submits remarks for task forces	High
61.			SRS_REQ_506	TC-10-06	TFMS notifies HOD/PSM after remarks submission	High
62.			SRS_REQ_507,508	TC-10-07	Lecturer accesses "My Portfolio" but has no task force assignments; TFMS displays "You have no task force assignments"	High
63.			SRS_REQ_509	TC-10-08	PDF download fails due to system error; TFMS displays error notification	High
64.	Executive Reporting	Executive Reports	SRS_REQ_601	TC-11-01	Management accesses Executive Reports section	High
65.			SRS_REQ_602	TC-11-02	TFMS displays Executive Reports dashboard with key metrics	High
66.			SRS_REQ_603	TC-11-03	Management drills down to department-level details	High
67.			SRS_REQ_604	TC-11-04-01	Management generates a report based on selected parameters: All Task force	High
68.			SRS_REQ_604	TC-11-04-02	Management generates a report based on selected parameters: Department-wise	High
69.			SRS_REQ_605	TC-11-05	TFMS exports report successfully in chosen format (e.g., PDF, Excel)	High